

Mahesh S

192412275

13. IMPLEMENT THE CAR PRICE PREDICTION MODEL USING PYTHON

1. Car Price Prediction – Linear Regression

Engine	Age	Price			
1000	8	3.0			
1200	7	3.8			
1500	5	5.0			
1800	3	6.5			
2000	2	7.5			

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("car1.csv")
X = df[["Engine", "Age"]]
y = df["Price"]
model = LinearRegression()
model.fit(X, y)
sample = [[1600, 4]]
print("Predicted Car Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Car Price: 5.7 Lakhs

2. Car Price Prediction – Random Forest

Year	KM Driven	Price
2015	80000	4.0
2016	70000	4.5
2018	50000	6.0
2020	30000	8.0
2022	15000	10.0

PROGRAM

```
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
df = pd.read_csv("car2.csv")
X = df[["Year", "KM Driven"]]
y = df["Price"]
model = RandomForestRegressor(
    n_estimators=50, random_state=1)
model.fit(X, y)
sample = [[2019, 40000]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 7.0 Lakhs

3. Car Price Prediction – Decision Tree

Mileage	Engine	Price
20	1000	4.0
18	1200	4.8
15	1500	6.0
12	1800	7.5
10	2000	9.0

PROGRAM

```
import pandas as pd
from sklearn.tree import DecisionTreeRegressor
df = pd.read_csv("car3.csv")
X = df[["Mileage", "Engine"]]
y = df["Price"]
model = DecisionTreeRegressor(random_state=2)
model.fit(X, y)
Sample = [[14, 1600]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 6.0 Lakhs

4. Car Price Prediction – Polynomial Regression

Car Age	Price
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1	10.0
2	8.5
3	7.2
4	6.0
5	5.0

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("car4.csv")
X = df[["Car Age"]]
y = df["Price"]
poly = PolynomialFeatures(degree=2)
Xp = poly.fit_transform(X)
model = LinearRegression()
model.fit(Xp, y)
sample = [[3.5]]
result = model.predict(poly.transform(sample))
print("Predicted Price:",
      round(result[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 6.55 Lakhs

5. Car Price Prediction – Multiple Linear Regression

Year	Engine	KM Driven	Price
2016	1200	70000	4.0
2017	1400	60000	4.8
2018	1500	50000	5.5
2020	1800	30000	7.5
2022	2000	15000	9.5

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("car5.csv")
X = df[["Year", "Engine", "KM Driven"]]
y = df["Price"]
model = LinearRegression()
model.fit(X, y)
sample = [[2019, 1600, 40000]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 6.5 Lakhs

